

Solution Requirements

Date	14 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should do).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Crop Recommendation	The system shall accept soil parameters (N, P, K, temperature, humidity, pH, rainfall) from users and recommend the most suitable crop using trained ML models.	High
2	User Input Interface	The system shall provide a responsive HTML web form for users to input agricultural parameters easily and receive instant crop prediction results.	High
3	External Interfaces	The system integrates with the agricultural CSV dataset and uses Scikit-learn ML library for model training. No external third-party APIs required; all processing is local.	Medium
4	Data Processing	The system shall preprocess input data including null value handling, outlier treatment, feature scaling, and seasonal crop extraction before passing it to the ML model.	High
5	Reporting & Results	The system shall display the predicted crop name along with model confidence score on a results page after processing the user-provided agricultural parameters.	High
6	Business Rules	All 7 parameters (N, P, K, temperature, humidity, pH, rainfall) must be provided by the user. The system shall validate inputs and reject incomplete or out-of-range values.	High
7	Compliance	The system shall use publicly available and ethically sourced agricultural datasets. All data used for training must be properly attributed and free from licensing violations.	Medium
8	Model Management	The system shall save the best-performing ML model in .pkl format using Pickle and load it automatically during each prediction request without retraining.	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The Flask web application shall process user inputs and return crop predictions within an acceptable response time.	Prediction response in < 2 seconds
2	Scalability	The system shall be designed to support larger agricultural datasets and additional ML models in future versions without major architectural changes.	Supports dataset scaling up to 100,000+ records
3	Security & Data Privacy	All user-submitted agricultural data shall be processed locally and not stored permanently. No personal data is collected or transmitted externally.	No external data transmission; local processing only
4	Reliability & Availability	The application shall run reliably on the Flask local development server without crashes during normal operation and testing.	99% uptime during local testing sessions
5	Usability & Accessibility	The web interface shall be simple, intuitive, and usable by farmers and non-technical users without prior training or technical knowledge.	Usable without training; Bootstrap-based responsive UI
6	Maintainability & Portability	The codebase shall follow modular Python coding standards with clear comments, making it easy to update ML models, add features, or deploy on cloud platforms in future.	Modular code; deployable on AWS/Heroku with minor changes